

Weakly Supervised Localisation Generalises Where Genotype Prediction Does Not: An Externally Validated Null for *EGFR* Radiogenomics from Lung CT

Manuscript type: Original Research

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Word counts.

- Abstract: [insert — limit 250]
- Manuscript, Introduction through Discussion: [insert — limit 3000]
- Figures: 6 (limit 6) Tables: 3 (limit 4) References: 15 (limit 35)

Summary statement. Weakly supervised tumour localisation transferred from a United States surgical cohort to a Dutch radiotherapy cohort, whereas prediction of epidermal growth factor receptor mutation status from the same lesion-anchored computed tomographic images did not exceed a five-variable model built from routinely recorded clinical data.

Institutional review board / ethics. [REQUIRED. Insert the board's written determination: formal approval, a waiver of informed consent, or an exempt / not-human-subjects-research determination, with its reference number and date. State HIPAA-compliance status if any author's institution is in the United States. The study analysed only publicly available, fully de-identified data distributed by The Cancer Imaging Archive under open licences, and no protected health information was accessed — but the board's determination, not the authors' reading of the regulations, is what the journal requires.]

Informed consent. [Insert. Expected: waived, as the analysis uses de-identified public data with no subject contact — confirm with the wording the board issues.]

Clinical trial registration. Not applicable. This is a retrospective secondary analysis of a public de-identified imaging archive, not a clinical trial. The pre-specified analysis plan is registered on the Open Science Framework: [OSF registration ID — REQUIRED before submission].

Funding. [REQUIRED. List every grant number and funder, or state: "The authors received no specific funding for this work." State the role of the funders in study design, data collection, analysis, and the decision to publish.]

Conflicts of interest. [REQUIRED for every author. Each must also complete the ICMJE disclosure form at submission.]

Data sharing statement. All analysis code, the pre-specified analysis plan with its deviation log, the completed CLAIM checklist, and every archived run result — including negative results — are publicly available at [permanent repository URL] and archived at [Zenodo DOI — REQUIRED]

before submission]. Every numerical result reported in this manuscript is regenerated from archived out-of-fold predictions by a single script, requiring no imaging data and no GPU.

Imaging data are not redistributed by the authors. Both collections are distributed by The Cancer Imaging Archive under their own terms: NSCLC-Radiogenomics under CC BY 3.0 and NSCLC-Radiomics under CC BY-NC 3.0. Only NSCLC-Radiogenomics images appear in any figure, consistent with those licences.

Previous presentation / preprint. [REQUIRED. State any prior presentation at a scientific meeting, and any posting to a preprint server, with the server, DOI and date. If none, state “This work has not been previously presented or posted to a preprint server.”]

Subject overlap with prior publications. The two imaging collections analysed here are public and have been analysed in prior work by other groups, including the collections’ source publications and a recent multi-tumour external-validation study in which NSCLC-Radiogenomics serves as the external cohort. No patient in this analysis overlaps with any prior publication by these authors. [Confirm and expand in the cover letter.]

Acknowledgments. [Insert here — must NOT appear in the anonymized main manuscript file.]

Submission checklist (delete before submitting).

- ☐ CLAIM checklist completed and uploaded
- ☐ Anonymized main manuscript contains no names, institutions, acknowledgments, or identifying URLs
- ☐ Repository link in the main file replaced by an anonymized mirror; real URL disclosed in the cover letter only
- ☐ Institutional review board determination obtained and inserted
- ☐ OSF registration ID and Zenodo DOI inserted
- ☐ Funding, conflicts, and ICMJE forms for every author
- ☐ At least two manuscript categorization terms selected in the submission system
- ☐ Statistical review sought — the journal advises consultation before analysis, and it has not yet happened